

Vijayaragavan S

SOFTWARE ENGINEER

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Computer Science undergraduate graduating in 2027, with experience building web applications, backend APIs and offline Android software. Work includes an AWS-hosted rental marketplace, a Java-based ML screening app and first-author research published in IEEE Xplore.

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, TypeScript, SQL, C, C++

Software development: React, Node.js, Express.js, Flask, REST APIs, Socket.IO, Android

Data and deployment: PostgreSQL, Prisma, AWS (EC2, S3, CloudFront, SES), Docker, Nginx, Git

Foundations and testing: Data structures and algorithms, object-oriented design, DBMS, pytest

PROJECTS

CampusRent | Campus Rental Marketplace

2025-Present | React, TypeScript, Node.js, PostgreSQL, AWS | Live application

- Built listings, bookings, reviews and real-time messaging, with separate student, landlord and admin access; modeled users and rental transactions in Prisma/PostgreSQL.
- Deployed the frontend on S3/CloudFront and the API on EC2 behind Nginx; integrated S3 image uploads and SES email verification.
- Implemented JWT authentication, configurable rate limits, request sanitization and a health endpoint reporting uptime and memory usage.

AnaemiaScreen | Offline Android Screening Prototype

2026 | Java, Python, PyTorch, LiteRT, OpenCV

- Built an offline Android app that takes a camera or gallery image through U-Net segmentation, image preprocessing and a calibrated ML ensemble.
- Separated image-processing logic from Android UI code for desktop validation; saved comparisons show zero decision differences across **634 scored** Python/Java cases.
- Implemented participant records, automatic image crops and CSV/ZIP export, with temporary-file writes and folder renames to protect incomplete records.

Real-Time Pothole Detection | Embedded ML

2025-Present | Python, C, TensorFlow Lite, ESP32 | Source code

- Built a dual-sensor detection pipeline using 3,245 labeled signal windows; the temporal convolutional model with attention achieved **96.7%** test accuracy.
- Compressed the model using distillation, pruning and quantization, retaining **95.8%** accuracy at **342 KB** and **47 ms** inference on ESP32.

PUBLICATION

Vijayaragavan Sivasamy et al. (first author). Real-Time Pothole Detection Using Dual Accelerometer Sensors with Temporal Convolutional Networks and Attention Mechanisms. IEEE conference publication, 2026.

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EDUCATION

B.Tech, Computer Science and Engineering | 2023-2027 (Expected)

SRM Institute of Science and Technology, Chennai | **CGPA: 9.46/10**

Relevant coursework: Data Structures and Algorithms, DBMS, Operating Systems, Computer Networks

Certifications: Java - NPTEL | SAP Backend ABAP Cloud Certificate